

# SOFTWARE REQUIREMENTS SPECIFICATION

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## LabPilot

Smart Classroom & Computer Lab Management System

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**Version:** 1.0

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# Chapter 1

## Introduction

### 1.1 Purpose

The purpose of this project is to develop a desktop-based Smart Classroom & Computer Lab Management System that enables instructors to monitor, manage, and coordinate computer laboratory sessions through a centralized interface.

The proposed system aims to improve classroom discipline, reduce manual supervision, simplify communication between teachers and students, and increase productivity during practical sessions by providing real-time monitoring and management tools over a Local Area Network (LAN).

### 1.2 Scope

The system is intended for:

- Engineering colleges
- Schools
- Universities
- Computer training institutes
- Coding bootcamps

The system will provide:

- Live classroom monitoring
- Student management
- Application management
- Task assignment
- Idle detection
- Notifications
- Attendance
- Activity logging

The system is designed to operate entirely within a Local Area Network without requiring internet connectivity.

## 1.3 Objectives

The primary objectives are:

- Centralize classroom management
- Improve teacher productivity
- Reduce student distractions
- Simplify task assignment
- Monitor student activities
- Detect idle students
- Maintain attendance automatically
- Provide activity reports
- Build an affordable alternative to commercial classroom management software

# Chapter 2

## Overall Description

### 2.1 Product Perspective

The proposed system consists of two desktop applications.

#### 2.1.1 Teacher Application

Acts as the administrator. Responsibilities include:

- Create sessions
- Monitor students
- Assign tasks
- Receive notifications
- Manage classroom

#### 2.1.2 Student Application

Runs on every student computer. Responsibilities include:

- Join classroom
- Share screen previews
- Report activity
- Receive tasks
- Receive teacher commands

### 2.2 Users

#### 2.2.1 Teacher

Responsibilities:

- Create classroom
- Manage students
- View screens
- Assign work
- Block applications
- Receive alerts
- Generate reports

### 2.2.2 Student

Responsibilities:

- Join classroom
- Complete assigned tasks
- Receive announcements
- Share activity status

## 2.3 Operating Environment

- **Operating System:** Windows 10, Windows 11
- **Network:** Local Area Network (LAN)
- **Desktop Platform:** Electron
- **Database:** SQLite

# Chapter 3

## System Features

Version 1 will include the following features.

### 3.1 Session Management

- Create session
- Join session
- End session

### 3.2 Student Management

- Student list
- Connection status
- Attendance

### 3.3 Screen Monitoring

- Live screen preview
- Full screen view
- Screenshot capture

### 3.4 Task Management

- Assign task (individual or broadcast assignment)
- Task status

### 3.5 Application Monitoring

- Running applications
- Active window
- Application history

### 3.6 Application Control

- Allow applications



- Block applications

### 3.7 Website Control (Basic)

- Allow list
- Block list

*Implementation approach to be finalized during the feasibility study.*

### 3.8 Idle Detection

- Mouse inactivity
- Keyboard inactivity
- Idle notification

### 3.9 Notifications

Examples include:

- Student joined
- Student disconnected
- Student idle
- Restricted application opened

### 3.10 Reports

- Attendance
- Activity log
- Session summary

# Chapter 4

## External Interface Requirements

### 4.1 User Interface

#### 4.1.1 Teacher Dashboard

Contains:

- Student grid
- Notifications
- Tasks
- Attendance
- Reports

#### 4.1.2 Student Interface

Contains:

- Session status
- Assigned task
- Connection status

### 4.2 Hardware Interface

- Teacher computer — Windows PC
- Student computers — Windows PCs
- LAN switch / Wi-Fi router

### 4.3 Software Interface

- Electron
- React
- TypeScript
- Node.js
- Express
- Socket.IO
- SQLite

## Chapter 5

# Functional Requirements

| ID    | Requirement                                          |
|-------|------------------------------------------------------|
| FR-01 | Teacher shall be able to create a classroom session. |
| FR-02 | Student shall be able to join using a session code.  |
| FR-03 | Teacher shall view all connected students.           |
| FR-04 | Teacher shall monitor student screens.               |
| FR-05 | Teacher shall assign tasks.                          |
| FR-06 | Teacher shall monitor running applications.          |
| FR-07 | Teacher shall block or allow applications.           |
| FR-08 | System shall detect idle students.                   |
| FR-09 | System shall generate notifications.                 |
| FR-10 | System shall maintain attendance.                    |
| FR-11 | System shall store activity logs.                    |
| FR-12 | System shall generate reports.                       |

# Chapter 6

## Non-Functional Requirements

### 6.1 Performance

- Screen updates within acceptable LAN latency.
- Support up to 30–50 student systems in Version 1.

### 6.2 Reliability

- Automatic reconnection after temporary network interruptions.

### 6.3 Security

- Session-based authentication.
- Secure communication between teacher and student applications.

### 6.4 Usability

- Intuitive interface requiring minimal training.

### 6.5 Scalability

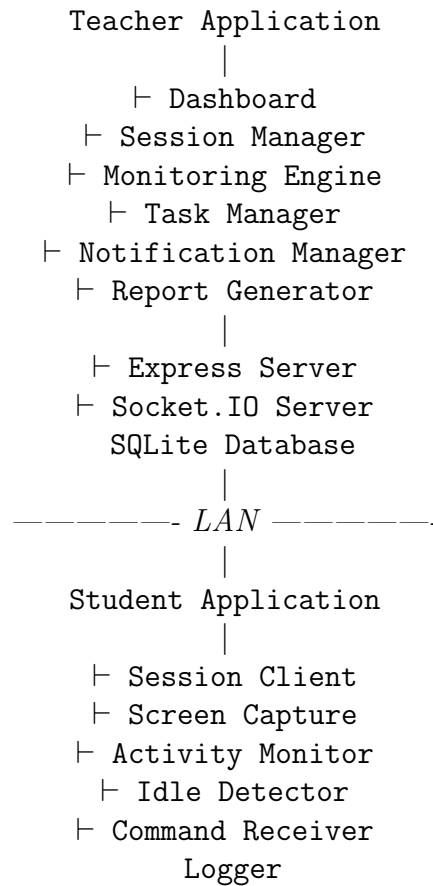
- Architecture should support future expansion to 100+ systems.

### 6.6 Maintainability

- Modular codebase.
- Reusable components.
- Clear project structure.

# Chapter 7

## System Architecture Overview



## Chapter 8

### Use Case Summary

*To be detailed in a subsequent revision of this document.*

# Chapter 9

## Constraints

- Windows operating system only (Version 1).
- Local Area Network required.
- No cloud dependency.
- Internet not required.
- Student application must be installed on each client computer.
- Some features (such as website control) may require administrator permissions or additional implementation mechanisms, depending on the chosen approach.

# Chapter 10

## Assumptions

- Teacher and student computers are connected to the same LAN.
- Student devices are institution-managed.
- The Teacher application starts the classroom session.
- The Student application is installed before the session begins.
- Users have the necessary permissions required by the deployed features.



# Chapter 11

## Future Enhancements

- AI-based classroom analytics
- Remote desktop control
- Teacher screen broadcasting
- File distribution
- Exam mode
- Multi-classroom management
- Mobile application
- Cloud synchronization
- Advanced analytics dashboard